

Anchoring and adjustment heuristic	: Heuristic whereby a person is heavily dependent on the information received initially (called the “anchor”) and then adjusts it towards the problem while making decisions.
Market Anomaly	: An anomaly is a deviation from the presently accepted paradigms or assumptions about the market which cannot be ignored as a random error.
Availability Heuristics	: It refers to the heuristic that assesses the probability of occurrence of the event based on the ease with which it comes to the mind.
Behavioural Finance	: It focuses on cognitive, sociological and psychological influences on the decision-making of investors and financial markets.
Cognitive Dissonance	: It refers to the internal tension and anxiety that people experience when they are subjected to conflicting beliefs or experiences opposite to their prior beliefs.
Disposition Effects	: It refers to the tendency to prematurely sell assets that have made financial gains while holding on to assets that are losing money.
Efficient Market Hypothesis	: An investment theory argues that markets are efficient, and prices reflect the fundamental value of an asset and thus there is no scope to make excess profits by investing.
Equity premium puzzle	: The equity premium puzzle refers to the inability of traditional finance models to explain the average equity risk premium over bonds.
Herding	: Herding is the outcome of social learning when people believe that others are better informed and thus follow the crowd.
Heuristics	: Heuristics or ‘rules of thumb’ are imperfect but functional shortcuts that help investors simplify complex situations and make quick decisions.
Loss Aversion	: Investors find losses to be more painful as compared to the pleasure from gains. Loss aversion makes people prefer things as they are.
Mental Accounting	: Mental accounting refers to a set of cognitive operations that investors use to classify their money differently in different portfolios based on various subjective criteria.
Predictability Puzzle	: Stock returns are predictable suggesting that persistent profits are not being eroded by arbitrage.

- Random Walk** : It refers to past movements or trends of a stock price or market that cannot be used to predict its future movement and it is not possible to outperform the market without assuming additional risk.
- Representativeness Heuristic** : In situations of uncertainty, people evaluate the probability of an uncertain event by comparing it to an existing prototype that already exists in our minds.
- Sunk Cost Fallacy** : It describes an emotional tendency to invest more money, time, and effort into a project where we have already invested even if a person knows that the investment is going to fail.
- Theory of Regret** : When an investment goes bad, people act irrationally by trying to avoid selling it to avoid feelings of regret.
- Volatility Puzzle** : The volatility puzzle reflects the high variability of stock prices having large dispersions
- Law of One Price** : The law says that identical securities have identical prices because prices are based on fundamental values. The law of one price limits the opportunity to arbitrage.

29.9 SOME USEFUL BOOKS

Baddeley, Michelle (2013). *Behavioural Economics and Finance*. Routledge. Chapter 11: Financial Instability

Ranyard, Rob(2018) *Economic Psychology*. British Psychological Society and Wiley. Chapter 15: Behaviour in Financial Markets.

Tversky, A., & Kahneman, D. (1986). Judgment under uncertainty: Heuristics and biases. In H. R. Arkes & K. R. Hammond (Eds.), *Judgment and decision making: An interdisciplinary reader* (pp. 38–55). Cambridge University Press. (This chapter originally appeared in "Science," 1974, 185, 1124-1131)

29.10 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) It focuses on cognitive, sociological and psychological influences on the decision-making of investors and financial markets.
- 2) In efficient markets, the price reflects its fundamental value and all the available information, that investors act rationally, have self-control and act in self-interest.

- 3) c
- 4) a

Check Your Progress 2

- 1) Traditional finance centres around the idea of investor rationality whereas Behavioural finance considers the influence of various biases in investment decision making.

Traditional finance assumes the availability of complete information, Behavioural finance, on the other hand, centres around the idea of bounded rationality.

Traditional finance centres around the 'Efficient market hypothesis'. Behavioural finance discusses various anomalies that lead to market volatility and cannot be explained through EMH.

Classical finance theory asserts fact that any noise is balanced out by rational traders but behavioural finance asserts that there is a limit to arbitrage and prices are not always right.

- 2) Refer to Section 1.4
- 3) B
- 4) A.

Check Your Progress 3

- 1) These are Fundamental anomalies, Technical anomalies, and Calendar or seasonal anomalies (refer to Sub-section 29.5.2).
- 2) Myopic loss aversion and ambiguity aversion explain the equity premium puzzle.
- 3) Various cognitive, social and emotional biases lead to sub-optimal decision-making by the investors which in turn affect the stock market and stock prices.
- 4) d

Check Your Progress 4

- 1) These are Over-Confidence, Cognitive dissonance, Under-reaction and over-reaction, Disposition effects, Diversification bias, and Mental Accounting (refer to Sub-section 29.6.2 for details).
- 2) The sunk cost fallacy describes an emotional tendency to invest more money, time, and effort into a project where we have already invested even if a person knows that the investment is going to fail.
- 3)
 - a) Herding
 - b) Self-attribution bias
 - c) Disposition effect
 - d) Loss aversion

29.11 TERMINAL QUESTIONS

- 1) How is behavioural finance different from traditional finance?
- 2) Behavioural Finance posits that heuristics help investors to make decisions quickly. Discuss the various heuristics and how are they applied by investors. Elaborate on the biases that these heuristics cause.
- 3) Give behavioural explanations to various market puzzles that are not addressed by traditional finance.
- 4) Describe various anomalies in the Efficient Markets Hypothesis.
- 5) Bring out the cognitive influences on investors that lead to various biases in decision-making.
- 6) Explain social influences on investors' decision-making
- 7) What are the various emotional influences that bias investor decision-making?
- 8) Define:
 - a) Sunk cost fallacy
 - b) Herding
 - c) Loss aversion
 - d) Mental accounting
 - e) Diversification bias
- 9) What is the Theory of Regret? How does it influence decision-making?
- 10) Describe the equity premium puzzle.
- 11) Describe predictability puzzle
- 12) 'Behavioural finance is 'micro efficient' but not macro efficient'. Comment.